JOINT ENTRANCE EXAMINATION (MAIN) JULY -2022
(CANDIDATE RESPONSE SHEET)

Paper/Subject	B.E/B.Tech.(Paper 1)
Exam Date	25 Jul 2022
Exam Slot	1

Question ID:100001
Topic Name: Mathematics-Section A
Question:

The total number of functions,
$f: \{1, 2, 3, 4\} \to \{1, 2, 3, 4, 5, 6\}$
such that $f(1)+f(2)=f(3)$, is equal to :

A 60

B 90

C 108

D 126

Answer Given By Candidate:**C**

Question ID:100002
Topic Name: Mathematics-Section A
Question:

If $\alpha, \beta, \gamma, \delta$ are the roots of the equation $x^4+x^3+x^2+x+1=0$, then $\alpha^{2021}+\beta^{2021}+\gamma^{2021}+\delta^{2021}$ is equal to :

A -4

B -1

C 1

D 4

Answer Given By Candidate:**Not Attempted**

Question ID:100003
Topic Name: Mathematics-Section A
Question:

For $n \in \mathbf{N}$, let $S_n = \left\{z \in \mathbf{C}: |z-3+2i| = \frac{n}{4}\right\}$ and $T_n = \left\{z \in \mathbf{C}: |z-2+3i| = \frac{1}{n}\right\}$. Then the number of elements in the set $\{n \in \mathbf{N} : S_n \cap T_n = \phi\}$ is :

A 0

B 2

C 3

D 4

Answer Given By Candidate:**D**

Question ID:100004
Topic Name: Mathematics-Section A

The number of $\theta \in (0, 4\pi)$ for which the system of linear equations

$$3(\sin 3\theta)\, x - y + z = 2$$
$$3(\cos 2\theta)\, x + 4y + 3z = 3$$
$$6x + 7y + 7z = 9$$

has no solution, is :

Question:

A 6

B 7

C 8

D 9

Answer Given By Candidate:**B**

Question ID:100005
Topic Name: Mathematics-Section A
Question:

If $\lim\limits_{n\to\infty} \left(\sqrt{n^2 - n - 1} + n\alpha + \beta\right) = 0$, then $8(\alpha + \beta)$ is equal to :

A 4

B -8

C -4

D 8

Answer Given By Candidate:**C**

Question ID:100006
Topic Name: Mathematics-Section A
Question:

If the absolute maximum value of the function $f(x) = (x^2 - 2x + 7)\, e^{(4x^3 - 12x^2 - 180x + 31)}$ in the interval $[-3, 0]$ is $f(\alpha)$, then :

A $\alpha = 0$

B $\alpha = -3$

C $\alpha \in (-1, 0)$

D $\alpha \in (-3, -1]$

Answer Given By Candidate:**Not Attempted**

Question ID:100007
Topic Name: Mathematics-Section A
Question:

The curve $y(x) = ax^3 + bx^2 + cx + 5$ touches the x-axis at the point $P(-2, 0)$ and cuts the y-axis at the point Q, where y' is equal to 3. Then the local maximum value of $y(x)$ is :

A $\frac{27}{4}$

B $\frac{29}{4}$

C $\frac{37}{4}$

D $\frac{9}{2}$

Answer Given By Candidate:**A**

Question ID:100008
Topic Name: Mathematics-Section A
Question: The area of the region given by $A = \{(x, y) : x^2 \le y \le \min\{x+2, 4-3x\}\}$ is :

A $\frac{31}{8}$

B $\frac{17}{6}$

C $\frac{19}{6}$

D $\frac{27}{8}$

Answer Given By Candidate:**Not Attempted**

Question ID:100009
Topic Name: Mathematics-Section A
Question:
For any real number x, let $[x]$ denote the largest integer less than equal to x. Let f be a real valued function defined on the interval $[-10, 10]$ by $f(x) = \begin{cases} x - [x], \text{ if } [x] \text{ is odd} \\ 1 + [x] - x, \text{ if } [x] \text{ is even} \end{cases}$.

Then the value of $\frac{\pi^2}{10} \int_{-10}^{10} f(x) \cos \pi x \, dx$ is :

A 4

B 2

C 1

D 0

Answer Given By Candidate:**Not Attempted**

Question ID:100010
Topic Name: Mathematics-Section A

Question:

The slope of the tangent to a curve C : $y = y(x)$ at any point (x, y) on it is $\frac{2e^{2x} - 6e^{-x} + 9}{2 + 9e^{-2x}}$.

If C passes through the points $\left(0, \frac{1}{2} + \frac{\pi}{2\sqrt{2}}\right)$ and $\left(\alpha, \frac{1}{2}e^{2\alpha}\right)$, then e^{α} is equal to :

A $\frac{3+\sqrt{2}}{3-\sqrt{2}}$

B $\frac{3}{\sqrt{2}}\left(\frac{3+\sqrt{2}}{3-\sqrt{2}}\right)$

C $\frac{1}{\sqrt{2}}\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right)$

D $\frac{\sqrt{2}+1}{\sqrt{2}-1}$

Answer Given By Candidate:**Not Attempted**

Question ID:100011
Topic Name: Mathematics-Section A
Question:

The general solution of the differential equation $(x - y^2)dx + y(5x + y^2)dy = 0$ is :

A $(y^2 + x)^4 = C|(y^2 + 2x)^3|$

B $(y^2 + 2x)^4 = C|(y^2 + x)^3|$

C $|(y^2 + x)^3| = C\ (2y^2 + x)^4$

D $|(y^2 + 2x)^3| = C\ (2y^2 + x)^4$

Answer Given By Candidate:**Not Attempted**

Question ID:100012
Topic Name: Mathematics-Section A
Question:

A line, with the slope greater than one, passes through the point A(4, 3) and intersects the line $x - y - 2 = 0$ at the point B. If the length of the line segment AB is $\frac{\sqrt{29}}{3}$, then B also lies on the line :

A $2x + y = 9$

B $3x - 2y = 7$

C $x + 2y = 6$

D $2x - 3y = 3$

Answer Given By Candidate:**Not Attempted**

Question ID:100013
Topic Name: Mathematics-Section A
Question:

Let the locus of the centre (α, β), $\beta > 0$, of the circle which touches the circle $x^2+(y-1)^2=1$ externally and also touches the x-axis be L. Then the area bounded by L and the line $y=4$ is :

A $\frac{32\sqrt{2}}{3}$

B $\frac{40\sqrt{2}}{3}$

C $\frac{64}{3}$

D $\frac{32}{3}$

Answer Given By Candidate:**C**

Question ID:100014
Topic Name: Mathematics-Section A
Question:

Let P be the plane containing the straight line $\frac{x-3}{9}=\frac{y+4}{-1}=\frac{z-7}{-5}$ and perpendicular to the plane containing the straight lines $\frac{x}{2}=\frac{y}{3}=\frac{z}{5}$ and $\frac{x}{3}=\frac{y}{7}=\frac{z}{8}$. If d is the distance of P from the point $(2, -5, 11)$, then d^2 is equal to :

A $\frac{147}{2}$

B 96

C $\frac{32}{3}$

D 54

Answer Given By Candidate:**C**

Question ID:100015
Topic Name: Mathematics-Section A

Question:

Let ABC be a triangle such that $\vec{BC} = \vec{a}$, $\vec{CA} = \vec{b}$, $\vec{AB} = \vec{c}$, $|\vec{a}| = 6\sqrt{2}$, $|\vec{b}| = 2\sqrt{3}$ and $\vec{b} \cdot \vec{c} = 12$. Consider the statements :

(S1) : $\left|(\vec{a} \times \vec{b}) + (\vec{c} \times \vec{b})\right| - |\vec{c}| = 6(2\sqrt{2} - 1)$

(S2) : $\angle ACB = \cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$

Then

A both (S1) and (S2) are true

B only (S1) is true

C only (S2) is true

D both (S1) and (S2) are false

Answer Given By Candidate:**Not Attempted**

Question ID:100016
Topic Name: Mathematics-Section A
Question:

If the sum and the product of mean and variance of a binomial distribution are 24 and 128 respectively, then the probability of one or two successes is :

A $\frac{33}{2^{32}}$

B $\frac{33}{2^{29}}$

C $\frac{33}{2^{28}}$

D $\frac{33}{2^{27}}$

Answer Given By Candidate:**C**

Question ID:100017
Topic Name: Mathematics-Section A
Question:

If the numbers appeared on the two throws of a fair six faced die are α and β, then the probability that $x^2 + \alpha x + \beta > 0$, for all $x \in \mathbf{R}$, is :

A $\frac{17}{36}$

B $\frac{4}{9}$

C $\frac{1}{2}$

D $\frac{19}{36}$

Answer Given By Candidate:**C**

Question ID:100018
Topic Name: Mathematics-Section A
Question: The number of solutions of $|\cos x| = \sin x$, such that $-4\pi \leq x \leq 4\pi$ is :

A 4

B 6

C 8

D 12

Answer Given By Candidate:**C**

Question ID:100019
Topic Name: Mathematics-Section A
Question:
A tower PQ stands on a horizontal ground with base Q on the ground. The point R divides the tower in two parts such that QR = 15 m. If from a point A on the ground the angle of elevation of R is 60° and the part PR of the tower subtends an angle of 15° at A, then the height of the tower is :

A $5(2\sqrt{3}+3)$ m

B $5(\sqrt{3}+3)$ m

C $10(\sqrt{3}+1)$ m

D $10(2\sqrt{3}+1)$ m

Answer Given By Candidate:**A**

Question ID:100020
Topic Name: Mathematics-Section A
Question: Which of the following statements is a tautology ?

A $((\sim p) \vee q) \Rightarrow p$

B $p \Rightarrow ((\sim p) \vee q)$

C $((\sim p) \vee q) \Rightarrow q$

D $q \Rightarrow ((\sim p) \vee q)$

Answer Given By Candidate:**D**

Question ID:100021
Topic Name: Mathematics-Section B

Question:

Let $A=\begin{pmatrix}2 & -1 & -1\\ 1 & 0 & -1\\ 1 & -1 & 0\end{pmatrix}$ and $B=A-I$. If $\omega=\frac{\sqrt{3}i-1}{2}$, then the number of elements in the set $\left\{n \in \{1,2,...,100\}: A^n+(\omega B)^n=A+B\right\}$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100022
Topic Name: Mathematics-Section B
Question:

The letters of the word 'MANKIND' are written in all possible orders and arranged in serial order as in an English dictionary. Then the serial number of the word 'MANKIND' is _____.

Answer Given By Candidate:**1492**

Question ID:100023
Topic Name: Mathematics-Section B
Question:

If the maximum value of the term independent of t in the expansion of $\left(t^2 x^{\frac{1}{5}}+\frac{(1-x)^{\frac{1}{10}}}{t}\right)^{15}$, $x \geq 0$, is K, then 8 K is equal to _____.

Answer Given By Candidate:**12012**

Question ID:100024
Topic Name: Mathematics-Section B
Question:

Let a, b be two non-zero real numbers. If p and r are the roots of the equation $x^2-8ax+2a=0$ and q and s are the roots of the equation $x^2+12bx+6b=0$, such that $\frac{1}{p},\frac{1}{q},\frac{1}{r},\frac{1}{s}$ are in A.P., then $a^{-1}-b^{-1}$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100025
Topic Name: Mathematics-Section B
Question:

Let $a_1=b_1=1$, $a_n=a_{n-1}+2$ and $b_n=a_n+b_{n-1}$ for every natural number $n \geq 2$. Then $\sum_{n=1}^{15} a_n \cdot b_n$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100026
Topic Name: Mathematics-Section B

Question:

Let $f(x) = \begin{cases} \left|4x^2 - 8x + 5\right|, & \text{if } 8x^2 - 6x + 1 \geq 0 \\ \left[4x^2 - 8x + 5\right], & \text{if } 8x^2 - 6x + 1 < 0, \end{cases}$ where $[\alpha]$ denotes the greatest integer less than or equal to α. Then the number of points in **R** where f is not differentiable is _____.

Answer Given By Candidate:**2**

Question ID:100027
Topic Name: Mathematics-Section B
Question:

If $\lim\limits_{n\to\infty} \dfrac{(n+1)^{k-1}}{n^{k+1}} \left[(nk+1) + (nk+2) + \ldots + (nk+n)\right]$

$= 33 \cdot \lim\limits_{n\to\infty} \dfrac{1}{n^{k+1}} \cdot \left[1^k + 2^k + 3^k + \ldots + n^k\right]$,

then the integral value of k is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100028
Topic Name: Mathematics-Section B
Question:

Let the equation of two diameters of a circle $x^2+y^2-2x+2fy+1=0$ be $2px-y=1$ and $2x+py=4p$. Then the slope $m\epsilon(0, \infty)$ of the tangent to the hyperbola $3x^2-y^2=3$ passing through the centre of the circle is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100029
Topic Name: Mathematics-Section B
Question:

The sum of diameters of the circles that touch (i) the parabola $75x^2=64(5y-3)$ at the point $\left(\frac{8}{5}, \frac{6}{5}\right)$ and (ii) the y-axis, is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100030
Topic Name: Mathematics-Section B
Question:

The line of shortest distance between the lines $\dfrac{x-2}{0} = \dfrac{y-1}{1} = \dfrac{z}{1}$ and $\dfrac{x-3}{2} = \dfrac{y-5}{2} = \dfrac{z-1}{1}$ makes an angle of $\cos^{-1}\left(\sqrt{\dfrac{2}{27}}\right)$ with the plane P : $ax-y-z=0$, $(a>0)$. If the image of the point $(1, 1, -5)$ in the plane P is (α, β, γ), then $\alpha+\beta-\gamma$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100031
Topic Name: Physics-Section A

Question:

If momentum [P], area [A] and time [T] are taken as fundamental quantities, then the dimensional formula for coefficient of viscosity is :

A $[P\,A^{-1}\,T^{0}]$

B $[P\,A\,T^{-1}]$

C $[P\,A^{-1}\,T]$

D $[P\,A^{-1}\,T^{-1}]$

Answer Given By Candidate:**A**

Question ID:100032
Topic Name: Physics-Section A
Question:

Which of the following physical quantities have the same dimensions ?

A Electric displacement ($\vec{D}$) and surface charge density

B Displacement current and electric field

C Current density and surface charge density

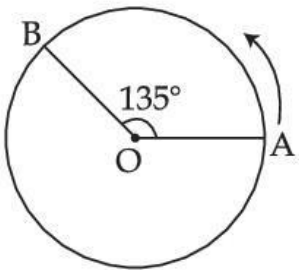
D Electric potential and energy

Answer Given By Candidate:**A**

Question ID:100033
Topic Name: Physics-Section A
Question:

A person moved from A to B on a circular path as shown in figure. If the distance travelled by him is 60 m, then the magnitude of displacement would be :

(Given $\cos 135° = -0.7$)



A 42 m

B 47 m

C 19 m

D 40 m

Answer Given By Candidate:**B**

Question ID:100034
Topic Name: Physics-Section A

Question:

If momentum [P], area [A] and time [T] are taken as fundamental quantities, then the dimensional formula for coefficient of viscosity is :

A 64 J

B 60 J

C 120 J

D 128 J

Answer Given By Candidate:**B**

Question ID:100035
Topic Name: Physics-Section A
Question:

A solid cylinder and a solid sphere, having same mass M and radius R, roll down the same inclined plane from top without slipping. They start from rest. The ratio of velocity of the solid cylinder to that of the solid sphere, with which they reach the ground, will be :

A $\sqrt{\frac{5}{3}}$

B $\sqrt{\frac{4}{5}}$

C $\sqrt{\frac{3}{5}}$

D $\sqrt{\frac{14}{15}}$

Answer Given By Candidate:**D**

Question ID:100036
Topic Name: Physics-Section A
Question:

Three identical particles A, B and C of mass 100 kg each are placed in a straight line with AB = BC = 13 m. The gravitational force on a fourth particle P of the same mass is F, when placed at a distance 13 m from the particle B on the perpendicular bisector of the line AC. The value of F will be approximately :

A 21 G

B 100 G

C 59 G

D 42 G

Answer Given By Candidate:**B**

Question ID:100037
Topic Name: Physics-Section A

Question:

A certain amount of gas of volume V at 27°C temperature and pressure $2 \times 10^7\ \text{Nm}^{-2}$ expands isothermally until its volume gets doubled. Later it expands adiabatically until its volume gets redoubled. The final pressure of the gas will be (Use $\gamma = 1.5$) :

A 3.536×10^5 Pa

B 3.536×10^6 Pa

C 1.25×10^6 Pa **D**

Answer Given By Candidate:**C**

1.25×10^5 Pa

Question ID:100038
Topic Name: Physics-Section A
Question:

Following statements are given :

(A) The average kinetic energy of a gas molecule decreases when the temperature is reduced.

(B) The average kinetic energy of a gas molecule increases with increase in pressure at constant temperature.

(C) The average kinetic energy of a gas molecule decreases with increase in volume.

(D) Pressure of a gas increases with increase in temperature at constant pressure.

(E) The volume of gas decreases with increase in temperature.

Choose the correct answer from the options given below :

A (A) and (D) only

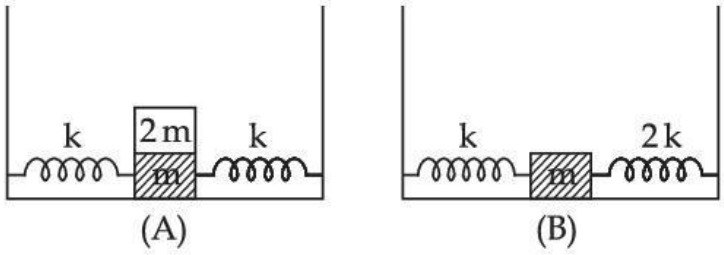
B (A), (B) and (D) only

C (B) and (D) only

D (A), (B) and (E) only

Answer Given By Candidate:**A**

Question ID:100039
Topic Name: Physics-Section A
Question:



In figure (A), mass '2 m' is fixed on mass 'm' which is attached to two springs of spring constant k. In figure (B), mass 'm' is attached to two springs of spring constant 'k' and '2k'. If mass 'm' in (A) and in (B) are displaced by distance 'x' horizontally and then released, then time period T_1 and T_2 corresponding to (A) and (B) respectively follow the relation.

A $\frac{T_1}{T_2} = \frac{3}{\sqrt{2}}$

B $\frac{T_1}{T_2} = \sqrt{\frac{3}{2}}$

C $\frac{T_1}{T_2} = \sqrt{\frac{2}{3}}$

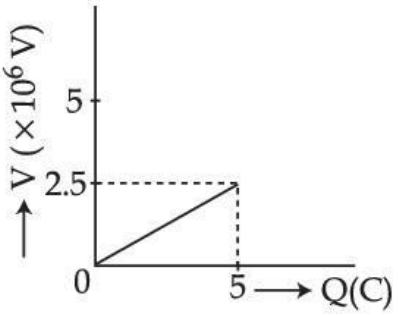
D $\frac{T_1}{T_2} = \frac{\sqrt{2}}{3}$

Answer Given By Candidate:**A**

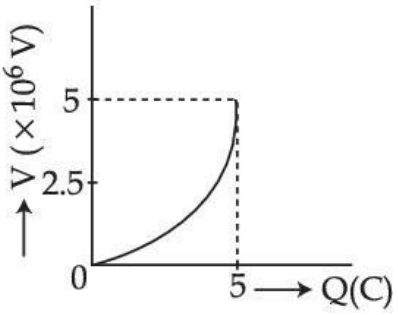
Question ID:100040
Topic Name: Physics-Section A
Question:

A condenser of 2 μF capacitance is charged steadily from 0 to 5 C. Which of the following graph represents correctly the variation of potential difference (V) across it's plates with respect to the charge (Q) on the condenser ?

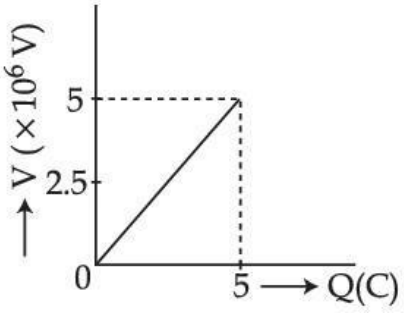
A



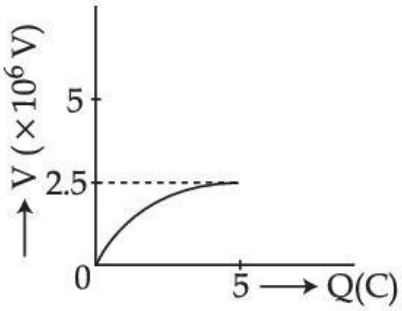
B



C



D



Answer Given By Candidate:**A**

Question ID:100041
Topic Name: Physics-Section A
Question:

Two charged particles, having same kinetic energy, are allowed to pass through a uniform magnetic field perpendicular to the direction of motion. If the ratio of radii of their circular paths is 6 : 5 and their respective masses ratio is 9 : 4. Then, the ratio of their charges will be :

A 8 : 5

B 5 : 4

C 5 : 3

D 8 : 7

Answer Given By Candidate:**B**

Question ID:100042
Topic Name: Physics-Section A
Question:

To increase the resonant frequency in series LCR circuit,

A source frequency should be increased.

B another resistance should be added in series with the first resistance.

C another capacitor should be added in series with the first capacitor.

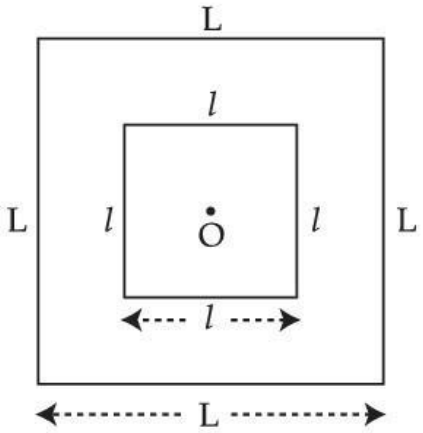
D the source frequency should be decreased.

Answer Given By Candidate:**Not Attempted**

Question ID:100043
Topic Name: Physics-Section A

Question:

A small square loop of wire of side l is placed inside a large square loop of wire L (L >> l). Both loops are coplanar and their centres coincide at point O as shown in figure. The mutual inductance of the system is :



A $\frac{2\sqrt{2}\,\mu_0 L^2}{\pi l}$

B $\frac{\mu_0 l^2}{2\sqrt{2}\,\pi L}$

C $\frac{2\sqrt{2}\,\mu_0 l^2}{\pi L}$

D $\frac{\mu_0 L^2}{2\sqrt{2}\,\pi l}$

Answer Given By Candidate:**C**

Question ID:100044
Topic Name: Physics-Section A
Question:

The rms value of conduction current in a parallel plate capacitor is 6.9 μA. The capacity of this capacitor, if it is connected to 230 V ac supply with an angular frequency of 600 rad/s, will be :

A 5 pF

B 50 pF

C 100 pF

D 200 pF

Answer Given By Candidate:**B**

Question ID:100045
Topic Name: Physics-Section A
Question:

Which of the following statement is correct ?

A In primary rainbow, observer sees red colour on the top and violet on the bottom

B In primary rainbow, observer sees violet colour on the top and red on the bottom

C In primary rainbow, light wave suffers total internal reflection twice before coming out of water drops.

D Primary rainbow is less bright than secondary rainbow.

Answer Given By Candidate:**A**

Question ID:100046
Topic Name: Physics-Section A
Question:

Time taken by light to travel in two different materials A and B of refractive indices μ_A and μ_B of same thickness is t_1 and t_2 respectively. If $t_2 - t_1 = 5 \times 10^{-10}$ s and the ratio of μ_A to μ_B is 1 : 2. Then, the thickness of material, in meter is : (Given v_A and v_B are velocities of light in A and B materials respectively.)

A $5 \times 10^{-10}\ v_A$ m

B 5×10^{-10} m

C 1.5×10^{-10} m

D $5 \times 10^{-10}\ v_B$ m

Answer Given By Candidate:**A**

Question ID:100047
Topic Name: Physics-Section A
Question:

A metal exposed to light of wavelength 800 nm and and emits photoelectrons with a certain kinetic energy. The maximum kinetic energy of photo-electron doubles when light of wavelength 500 nm is used. The workfunction of the metal is : (Take hc = 1230 eV-nm).

A 1.537 eV

B 2.46 eV

C 0.615 eV

D 1.23 eV

Answer Given By Candidate:**C**

Question ID:100048
Topic Name: Physics-Section A
Question:

The momentum of an electron revolving in n^{th} orbit is given by :
(Symbols have their usual meanings)

A $\frac{nh}{2\pi r}$

B $\frac{nh}{2r}$

C

$\frac{nh}{2\pi}$

D

$\frac{2\pi r}{nh}$

Answer Given By Candidate:**C**

Question ID:100049
Topic Name: Physics-Section A
Question:

The magnetic moment of an electron (e) revolving in an orbit around nucleus with an orbital angular momentum is given by :

A

$\vec{\mu}_L = \frac{\vec{eL}}{2m}$

B

$\vec{\mu}_L = -\frac{\vec{eL}}{2m}$

C

$\vec{\mu}_l = -\frac{\vec{eL}}{m}$

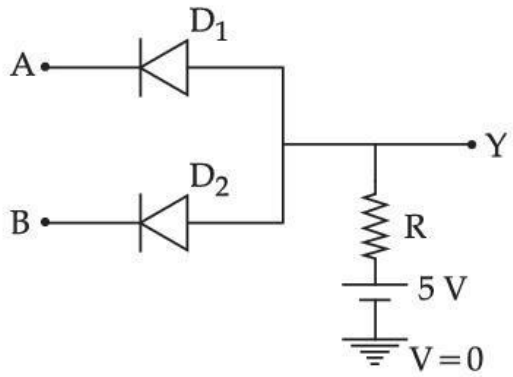
D

$\vec{\mu}_l = \frac{2\vec{eL}}{m}$

Answer Given By Candidate:**B**

Question ID:100050
Topic Name: Physics-Section A
Question:

In the circuit, the logical value of A = 1 or B = 1 when potential at A or B is 5 V and the logical value of A = 0 or B = 0 when potential at A or B is 0 V.



The truth table of the given circuit will be :

A

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	1

B

A	B	Y
0	0	0
1	0	1
0	1	1
1	1	1

C

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	0

D

A	B	Y
0	0	1
1	0	1
0	1	1
1	1	0

Answer Given By Candidate:**A**

Question ID:100051
Topic Name: Physics-Section B
Question:

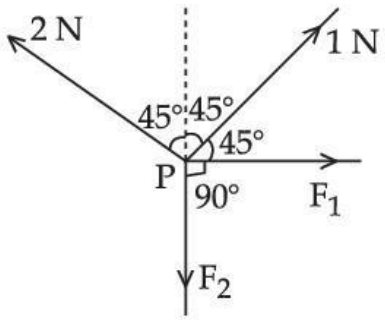
A car is moving with speed of 150 km/h and after applying the break it will move 27 m before it stops. If the same car is moving with a speed of one third the reported speed then it will stop after travelling _____ m distance.

Answer Given By Candidate:**3**

Question ID:100052
Topic Name: Physics-Section B

Question:

Four forces are acting at a point P in equilibrium as shown in figure. The ratio of force F_1 to F_2 is $1 : x$ where $x =$ _____.



Answer Given By Candidate:**3**

Question ID:100053
Topic Name: Physics-Section B
Question:

A wire of length L and radius r is clamped rigidly at one end. When the other end of the wire is pulled by a force F, its length increases by 5 cm. Another wire of the same material of length 4L and radius 4r is pulled by a force 4F under same conditions. The increase in length of this wire is _____ cm.

Answer Given By Candidate:**Not Attempted**

Question ID:100054
Topic Name: Physics-Section B
Question:

A unit scale is to be prepared whose length does not change with temperature and remains 20 cm, using a bimetallic strip made of brass and iron each of different length. The length of both components would change in such a way that difference between their lengths remains constant. If length of brass is 40 cm and length of iron will be _____ cm.

($\alpha_{iron} = 1.2 \times 10^{-5}\ K^{-1}$ and $\alpha_{brass} = 1.8 \times 10^{-5}\ K^{-1}$).

Answer Given By Candidate:**Not Attempted**

Question ID:100055
Topic Name: Physics-Section B
Question:

An observer is riding on a bicycle and moving towards a hill at 18 kmh^{-1}. He hears a sound from a source at some distance behind him directly as well as after its reflection from the hill. If the original frequency of the sound as emitted by source is 640 Hz and velocity of the sound in air is 320 m/s, the beat frequency between the two sounds heard by observer will be _____ Hz.

Answer Given By Candidate:**Not Attempted**

Question ID:100056
Topic Name: Physics-Section B

Question:

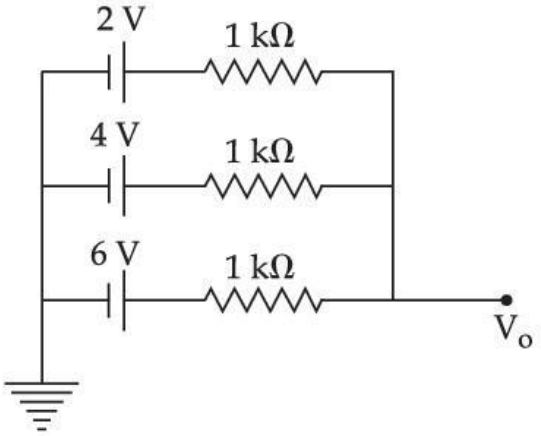
The volume charge density of a sphere of radius 6 m is 2 $\mu C\ cm^{-3}$. The number of lines of force per unit surface area coming out from the surface of the sphere is _____ $\times 10^{10}\ NC^{-1}$.

[Given : Permittivity of vacuum $\epsilon_0 = 8.85 \times 10^{-12}\ C^2\ N^{-1} - m^{-2}$)

Answer Given By Candidate:**Not Attempted**

Question ID:100057
Topic **Name:**Physics-Section B

In the given figure, the value of V_o will be _____ V.



Question:
Answer Given By Candidate:**4**

Question ID:100058
Topic Name: Physics-Section B
Question:

Eight copper wire of length *l* and diameter d are joined in parallel to form a single composite conductor of resistance R. If a single copper wire of length 2*l* have the same resistance (R) then its diameter will be _____d.

Answer Given By Candidate:**4**

Question ID:100059
Topic Name: Physics-Section B
Question:

The energy band gap of semiconducting material to produce violet (wavelength = 4000 Å) LED is _____ eV. (Round off to the nearest integer).

Answer Given By Candidate:**3**

Question ID:100060
Topic Name: Physics-Section B
Question:

The required height of a TV tower which can cover the population of 6.03 lakh is h. If the average population density is 100 per square km and the radius of earth is 6400 km, then the value of h will be _____ m.

Answer Given By Candidate:**Not Attempted**

Question ID:100061
Topic Name: Chemistry-Section A

Question:

SO_2Cl_2 on reaction with excess of water results into acidic mixture

$$SO_2Cl_2 + 2H_2O \rightarrow H_2SO_4 + 2HCl$$

16 moles of NaOH is required for the complete neutralisation of the resultant acidic mixture. The number of moles of SO_2Cl_2 used is :

A 16

B 8

C 4 **D**

Answer Given By Candidate:**C**

2

Question ID:100062
Topic Name: Chemistry-Section A
Question:

Which of the following sets of quantum numbers is not allowed ?

A

$$n=3,\ l=2,\ m_l=0,\ s=+\frac{1}{2}$$

B

$$n=3,\ l=2,\ m_l=-2,\ s=+\frac{1}{2}$$

C

$$n=3,\ l=3,\ m_l=-3,\ s=-\frac{1}{2}$$

D

$$n=3,\ l=0,\ m_l=0,\ s=-\frac{1}{2}$$

Answer Given By Candidate:**C**

Question ID:100063
Topic Name: Chemistry-Section A
Question:

The depression in freezing point observed for a formic acid solution of concentration 0.5 mL L^{-1} is 0.0405°C. Density of formic acid is 1.05 g mL^{-1}. The Van't Hoff factor of the formic acid solution is nearly : (Given for water $k_f = 1.86$ k kg mol^{-1})

A 0.8

B 1.1

C 1.9

D 2.4

Answer Given By Candidate:**Not Attempted**

Question ID:100064
Topic Name: Chemistry-Section A

Question:

20 mL of 0.1 M NH_4OH is mixed with 40 mL of 0.05 M HCl. The pH of the mixture is nearest to :

(Given : $K_b(NH_4OH) = 1 \times 10^{-5}$, log 2 = 0.30, log 3 = 0.48, log 5 = 0.69, log 7 = 0.84, log 11 = 1.04)

A 3.2

B 4.2

C 5.2

D 6.2

Answer Given By Candidate:**Not Attempted**

Question ID:100065
Topic Name: Chemistry-Section A

Match **List - I** with **List - II**

	List - I		**List - II**
(A)	$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$	(I)	Cu
(B)	$CO(g) + 3H_2(g) \rightarrow CH_4(g) + H_2O(g)$	(II)	$Cu/ZnO - Cr_2O_3$
(C)	$CO(g) + H_2(g) \rightarrow HCHO(g)$	(III)	$Fe_xO_y + K_2O + Al_2O_3$
(D)	$CO(g) + 2H_2(g) \rightarrow CH_3OH(g)$	(IV)	Ni

Choose the **correct** answer from the options given below :

Question:

A (A) - (II), (B) - (IV), (C) - (I), (D) - (III)

B (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

C (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

D (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

Answer Given By Candidate:**C**

Question ID:100066
Topic Name: Chemistry-Section A
Question:

The IUPAC nomenclature of an element with electronic configuration [Rn] $5f^{14}6d^{1}7s^{2}$ is :

A Unnilbium

B Unnilunium

C Unnilquadium

D Unniltrium

Answer Given By Candidate:**D**

Question ID:100067
Topic Name: Chemistry-Section A

The compound(s) that is(are) removed as slag during the extraction of copper is :

(A) CaO

(B) FeO

(C) Al_2O_3

(D) ZnO

(E) NiO

Question: Choose the **correct** answer from the options given below :

A (C), (D) only

B (A), (B), (E) only

C (A), (B) only

D (B) only

Answer Given By Candidate:**Not Attempted**

Question ID:100068
Topic Name: Chemistry-Section A
Question:

The reaction of H_2O_2 with potassium permanganate in acidic medium leads to the formation of mainly :

A Mn^{2+}

B Mn^{4+}

C Mn^{3+}

D Mn^{6+}

Answer Given By Candidate:**A**

Question ID:100069
Topic Name: Chemistry-Section A

Question: Choose the **correct** order of density of the alkali metals :

A Li < K < Na < Rb < Cs

B Li < Na < K < Rb < Cs

C Cs < Rb < K < Na < Li

D Li < Na < K < Cs < Rb

Answer Given By Candidate:**Not Attempted**

Question ID:100070
Topic Name: Chemistry-Section A

Question:

The geometry around boron in the product 'B' formed from the following reaction is

$$BF_3 + NaH \xrightarrow{450\ K} A + NaF$$
$$A + NMe_3 \rightarrow B$$

A trigonal planar

B tetrahedral

C pyramidal

D square planar

Answer Given By Candidate:**Not Attempted**

Question ID:100071
Topic Name: Chemistry-Section A
Question:

The interhalogen compound formed from the reaction of bromine with excess of fluorine is a :

A hypohalite

B halate

C perhalate

D halite

Answer Given By Candidate:**B**

Question ID:100072
Topic Name: Chemistry-Section A
Question:

The photochemical smog does not generally contain :

A NO

B NO_2

C SO_2

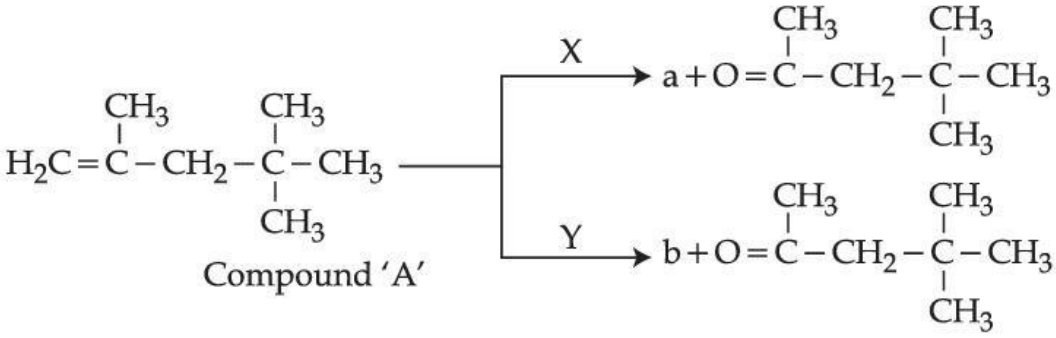
D HCHO

Answer Given By Candidate:**C**

Question ID:100073
Topic Name: Chemistry-Section A

Question:

A compound 'A' on reaction with 'X' and 'Y' produces the same major product but different by product 'a' and 'b'. Oxidation of 'a' gives a substance produced by ants.



'X' and **'Y'** respectively are

A $KMnO_4/H^+$ and dil. $KMnO_4$, 273 K

B $KMnO_4$(dilute), 273 K and $KMnO_4/H^+$

C $KMnO_4/H^+$ and O_3, H_2O/Zn

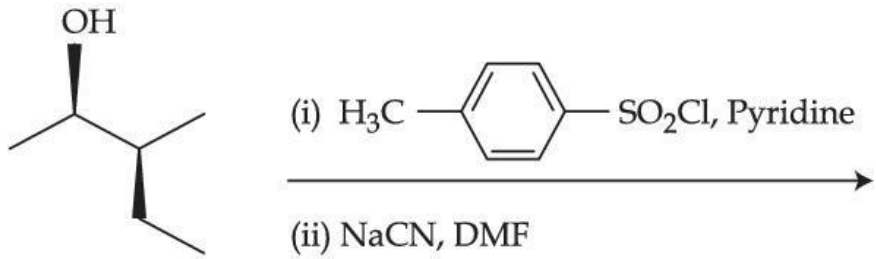
D O_3, H_2O/Zn and $KMnO_4/H^+$

Answer Given By Candidate:**D**

Question ID:100074

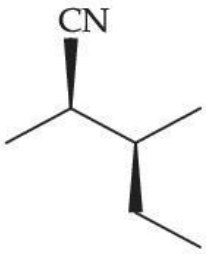
Topic **Name:**Chemistry-Section A

Most stable product of the following reaction is :

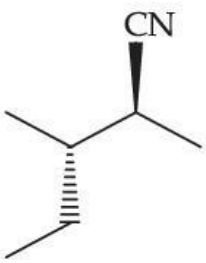


Question:

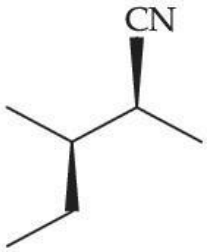
A



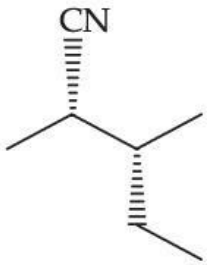
B



C



D

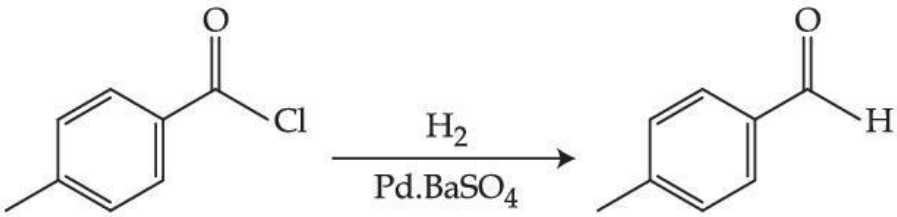


Answer Given By Candidate:**B**

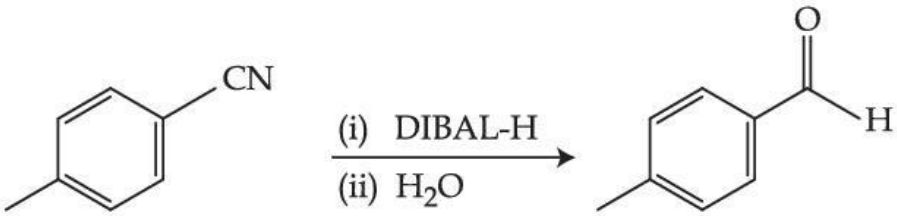
Question ID:100075
Topic Name: Chemistry-Section A
Question:

Which one of the following reactions does **not** represent correct combination of substrate and product under the given conditions ?

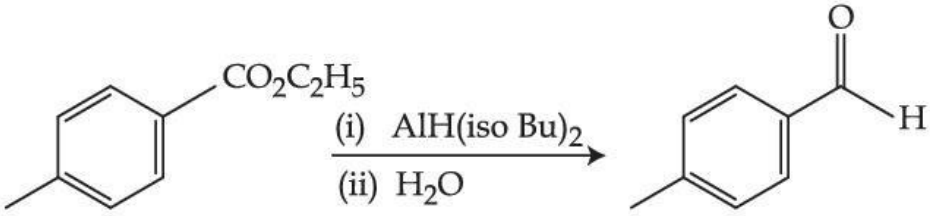
A



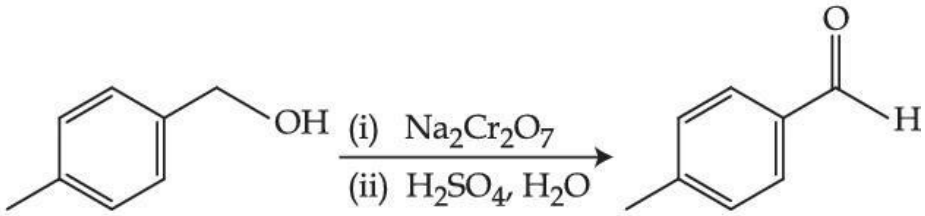
B



C



D



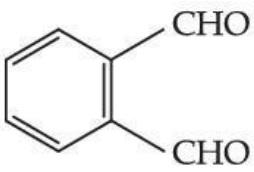
Answer Given By Candidate:**B**

Question ID:100076
Topic Name: Chemistry-Section A

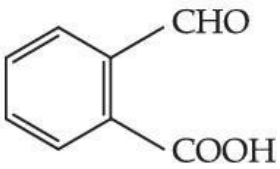
Question:

An organic compound 'A' on reaction with NH_3 followed by heating gives compound B. Which on further strong heating gives compound C ($C_8H_5NO_2$). Compound C on sequential reaction with ethanolic KOH, alkyl chloride and hydrolysis with alkali gives a primary amine. The compound A is :

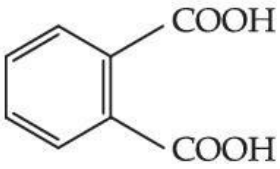
A



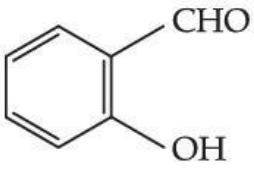
B



C



D

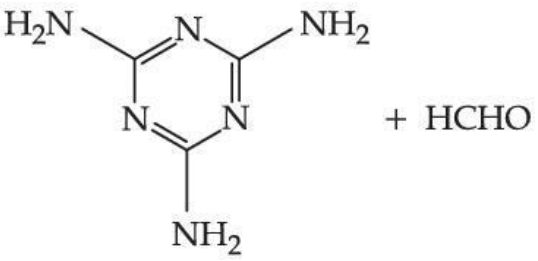


Answer Given By Candidate:**C**

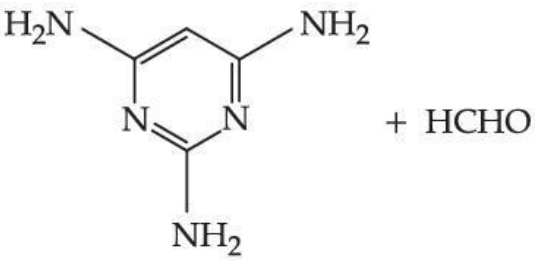
Question ID:100077
Topic Name: Chemistry-Section A

Question: Melamine polymer is formed by the condensation of :

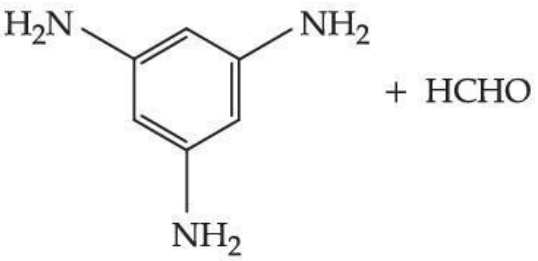
A



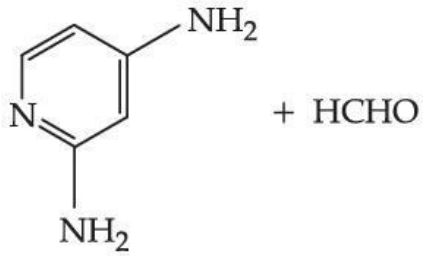
B



C



D



Answer Given By Candidate:**A**

Question ID:100078
Topic Name: Chemistry-Section A
Question: During the denaturation of proteins, which of these structures will remain intact ?

A Primary

B Secondary

C Tertiary

D Quaternary

Answer Given By Candidate:**A**

Question ID:100079
Topic Name: Chemistry-Section A
Question:

Drugs used to bind to receptors, inhibiting its natural function and blocking a message are called :

A Agonists

B Antagonists

C Allosterists

D Anti histaminists

Answer Given By Candidate:**B**

Question ID:100080
Topic Name: Chemistry-Section A
Question:

Given below are two statements :

Statement I : On heating with $KHSO_4$, glycerol is dehydrated and acrolein is formed.

Statement II : Acrolein has fruity odour and can be used to test glycerol's presence.

Choose the **correct** option.

A Both **Statement I** and **Statement II** are correct.

B Both **Statement I** and **Statement II** are incorrect.

C **Statement I** is correct but **Statement II** is incorrect.

D **Statement I** is incorrect but **Statement II** is correct.

Answer Given By Candidate:**Not Attempted**

Question ID:100081
Topic Name: Chemistry-Section B
Question:

Among the following species

$N_2, N_2^+, N_2^-, N_2^{2-}, O_2, O_2^+, O_2^-, O_2^{2-}$

the number of species showing diamagnesim is _____ .

Answer Given By Candidate:**2**

Question ID:100082
Topic Name: Chemistry-Section B
Question:

The enthalpy of combustion of propane, graphite and dihydrogen at 298 K are -2220.0 kJ mol^{-1}, -393.5 kJ mol^{-1} and -285.8 kJ mol^{-1} respectively. The magnitude of enthalpy of formation of propane (C_3H_8) is _____ kJ mol^{-1}. (Nearest integer)

Answer Given By Candidate:**Not Attempted**

Question ID:100083
Topic Name: Chemistry-Section B
Question:

The pressure of a moist gas at 27°C is 4 atm. The volume of the container is doubled at the same temperature. The new pressure of the moist gas is _____ $\times 10^{-1}$ atm. (Nearest integer)

(Given : The vapour pressure of water at 27°C is 0.4 atm.)

Answer Given By Candidate:**Not Attempted**

Question ID:100084
Topic Name: Chemistry-Section B
Question:

The cell potential for $Zn|Zn^{2+}(aq)||Sn^{x+}|Sn$ is 0.801 V at 298 K. The reaction quotient for the above reaction is 10^{-2}. The number of electrons involved in the given electrochemical cell reaction is _____.

(Given : $E^o_{Zn^{2+}|Zn} = -0.763$ V, $E^o_{Sn^{x+}|Sn} = +0.008$ V and $\frac{2.303RT}{F} = 0.06$ V)

Answer Given By Candidate:**4**

Question ID:100085
Topic Name: Chemistry-Section B
Question:

The half life for the decomposition of gaseous compound A is 240 s when the gaseous pressure was 500 Torr initially. When the pressure was 250 Torr, the half life was found to be 4.0 min. The order of the reaction is _____. (Nearest integer)

Answer Given By Candidate:**1**

Question ID:100086
Topic Name: Chemistry-Section B

Question:

Consider the following metal complexes :

$[Co(NH_3)_6]^{3+}$

$[CoCl(NH_3)_5]^{2+}$

$[Co(CN)_6]^{3-}$

$[Co(NH_3)_5(H_2O)]^{3+}$

The spin-only magnetic moment value of the complex that absorbes light with shortest wavelength is _____ B.M. (Nearest integer)

Answer Given By Candidate:**0**

Question ID:100087
Topic Name: Chemistry-Section B
Question:

Among Co^{3+}, Ti^{2+}, V^{2+} and Cr^{2+} ions, one if used as a reagent cannot liberate H_2 from dilute mineral acid solution, its spin-only magnetic moment in gaseous state is _____ B.M. (Nearest integer)

Answer Given By Candidate:**Not Attempted**

Question ID:100088
Topic Name: Chemistry-Section B
Question:

While estimating the nitrogen present in an organic compound by Kjeldahl's method, the ammonia evolved from 0.25 g of the compound neutralized 2.5 mL of 2 M H_2SO_4. The percentage of nitrogen present in organic compound is _____.

Answer Given By Candidate:**Not Attempted**

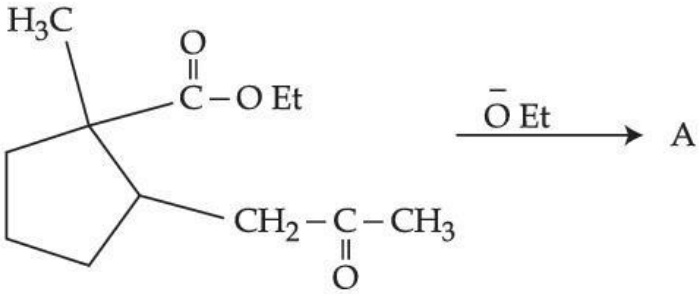
Question ID:100089
Topic Name: Chemistry-Section B
Question:

The number of sp^3 hybridised carbons in an acyclic neutral compound with molecular formula C_4H_5N is _____.

Answer Given By Candidate:**0**

Question ID:100090
Topic Name: Chemistry-Section B

In the given reaction



(Where Et is $-C_2H_5$)

Question:

The number of chiral carbon/s in product **A** is _____.

Answer Given By Candidate:**Not Attempted**